Interim Report: Predicting Price Moves with News Sentiment

Project: Financial News Sentiment Analysis and Stock Price Correlation

Challenge: Week 1 - Stock Challenge (Enhanced in Week 12)

Date Started: November 2024

Last Updated: February 17, 2026

Status: [DONE] Week 12 Improvements Completed

Week 12 Enhancement Summary

Improvements Completed

Category	Improvement	Status
Code Quality	Refactored to m	[DONE] Complete
Code Quality	Created datacla	[DONE] Complete
Testing	Added 35 unit t	[DONE] Complete
CI/CD	Enhanced GitHub	[DONE] Complete
Visualization	Built interacti	[DONE] Complete
Documentation	Professional RE	[DONE] Complete
Documentation	Updated reports	[DONE] Complete

New Project Structure

```
src/
 config.py           # Dataclasses and constants
 data/
    loader.py       # Data loading with type hints
    preprocessor.py # Preprocessing functions
 analysis/
    sentiment.py    # TextBlob and VADER sentiment
    technical.py    # RSI, MACD, SMA, EMA, etc.
    statistics.py   # Gini coefficient, correlations
 visualization/
     plotting.py     # Plotting utilities
```

Test Results

```
============================= 35 passed in 13.91s ==============================
```

Key Files Added

- app/dashboard.py - Streamlit interactive dashboard

Executive Summary

This report documents the progress of analyzing financial news sentiment and its correlation with stock market movements. The project aims to enhance predictive analytics capabilities for Nova Financial Solutions by establishing statistical relationships between news sentiment and stock price fluctuations.

Current Status: [DONE] All Week 1 tasks completed successfully!

- Task 1 EDA: Comprehensive exploratory data analysis of 1.4M+ headlines across 7 structured notebooks with 22 visualizations

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Project Overview

Business Objective

Nova Financial Solutions aims to enhance its predictive analytics capabilities to significantly boost financial forecasting accuracy and operational efficiency through advanced data analysis. The primary objectives are:

1. Sentiment Analysis: Perform sentiment analysis on headline text to quantify the tone and sentiment expressed in financial news using NLP techniques.

Dataset Overview

Financial News and Stock Price Integration Dataset (FNSPID)

The dataset structure includes:

- headline: Article release headline (title of news article)

Key Dates

- Challenge Introduction: 10:30 AM UTC, Wednesday, 19 Nov 2025

Progress Tracking

Task 1: Git and GitHub Setup & EDA

Status: [DONE] Completed

Branch: task-1 (merged to development)

Last Updated: December 2024

Completed Items

- Created GitHub repository

Descriptive Statistics

Headline Length Analysis:

- Calculated basic statistics (mean, median, std dev, skewness, kurtosis)

Text Analysis (Topic Modeling)

- Extracted common keywords/phrases

Time Series Analysis

- Analyzed publication frequency over time (hourly, daily, weekly, monthly)

Publisher Analysis

- Identified top contributing publishers

Git Activity

Commits Made:
- Initial project setup

Task 2: Quantitative Analysis using PyNance and TA-Lib

Status: [DONE] Completed
Branch: task-2 (merged to development)
Last Updated: December 2024

Completed Items

- Created task-2 branch

Technical Indicators

Moving Averages:
- Simple Moving Average (SMA) - Calculated for 20, 50, and 200-day periods

Financial Metrics

- Calculated daily and cumulative returns

Visualizations

- Stock price charts with moving averages

Task 3: Correlation between News and Stock Movement

Status: [DONE] Completed
Branch: task-3
Last Updated: December 2024

Completed Items

- Created task-3 branch

Data Preparation

Date Alignment:

- Normalized timestamps and extracted date-only values

Sentiment Analysis

Tools Used:

- NLTK (for text preprocessing)

Stock Movement Calculation

Daily Returns:

- Calculated percentage change in closing prices for all 6 stocks

Correlation Analysis

Aggregation:

- Computed average daily sentiment scores (43,466 date-stock combinations)

Key Findings & Insights

Preliminary Insights

1. Dataset Scale: The dataset contains over 1.4 million financial news headlines, providing a substantial foundation for analysis.

Challenges Encountered

1. Date Parsing Issues: The majority of dates (96.02%) in the dataset are invalid or improperly formatted. This required implementing robust date handling in the preprocessing pipeline to convert dates to datetime format and handle invalid entries gracefully.

Lessons Learned

1. Modular Design: Creating reusable utility functions (utils.py) early in the project significantly improved code organization and reusability across multiple notebooks.

Next Steps

Immediate Actions

1. Complete descriptive statistics analysis (headline length, publisher activity, publication trends)

Upcoming Tasks

- Merge Task 1 branch to main via PR

Technical Details

Environment Setup

- Python Version: [To be updated]

Data Sources

- News Data: data/raw_analyst_ratings.csv (1,407,328 rows)

Tools & Libraries

- Data Processing: pandas, numpy

References & Resources

[Add references, tutorials, and resources you've used]

1. [Resource 1]

Appendix

Code Snippets

[Add important code snippets or functions you've created]

Visualizations

Descriptive Statistics Visualizations:

1. Headline Length Analysis (notebooks/figures/headline_length_analysis.png)

Report Maintenance: This document should be updated regularly as you progress through the tasks. Commit changes with descriptive messages like "Update interim report: Task 1 EDA completed" or "Update interim report: Added correlation findings".